# Chapter 2. The Method of Mathematical Induction

*One Step at a Time*

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## 2.1 Weak induction

**Motivation.** To prove a statement for every natural number it suffices to check the first case and show that each case implies the next. Induction turns one infinite task into two finite ones; its price is that the statement to be proved must already be known or guessed.

**The method.** To prove a statement $S(m)$ for all natural $m$:

1. *Initial step:* prove $S(m)$ for the smallest $m$ for which it is defined (usually 0 or 1).
2. *Induction step:* prove that $S(n)$ (the *induction hypothesis*) implies $S(n+1)$.

*Validity.* If both steps are done but $S$ fails somewhere, let $m+1$ be the smallest failure. By the initial step, $m+1$ is not the first index, so $S(m)$ is defined and true by minimality; the induction step with $n = m$ then gives $S(m+1)$, a contradiction.

**Example 2.1**

For all positive integers $m$,

$$1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}. \tag{2.1}$$

*Solution.* For $m = 1$ both sides are 1. Assume (2.1) for $n$. The statement for $n+1$ differs from it by adding $(n+1)^2$ on the left and

$$\frac{(n+1)(n+2)(2n+3) - n(n+1)(2n+1)}{6} = \frac{(n+1)\big[(n+2)(2n+3) - n(2n+1)\big]}{6} = (n+1)^2$$

on the right. Adding this identity to (2.1) for $n$ gives the statement for $n+1$.

**Example 2.2**

Let $f(m)$ be the maximum number of regions into which $m$ straight lines divide the plane. Then $f(m) = \frac{m(m+1)}{2} + 1$.
*Solution.* $f(1) = 2$. Given $n+1$ lines, look at the last line $s$. It crosses $k \le n$ of the other lines, so it passes through $k+1$ regions formed by the others and cuts each of them in two. Hence $f(n+1) \le f(n)+n+1$, with equality when $s$ meets all other lines in distinct new points (lines in general position). Thus $f(n+1) = f(n)+n+1 = \frac{n(n+1)}{2}+1+n+1 = \frac{(n+1)(n+2)}{2}+1$.

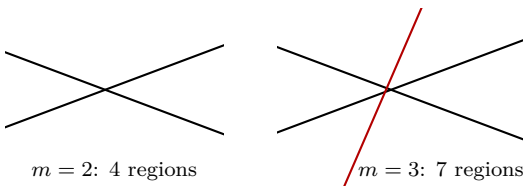



**Figure.** *Example 2.2: the new (red) line crosses both old lines, passes through 3 old regions and splits each, so* $f(3) = f(2) + 3$.

The formula had to be known in advance; finding it is the weak point of induction. Often it can be guessed from small cases:

**Example 2.3**

Let $a_0 = 1$ and $a_{n+1} = 3a_n + 1$. Then $a_m = \frac{3^{m+1}-1}{2}$.
*Solution.* The first values $1, 4, 13, 40, 121$ suggest the formula. It holds for $m = 0$, and if it holds for $n$, then $a_{n+1} = 3 \cdot \frac{3^{n+1}-1}{2} + 1 = \frac{3^{n+2}-1}{2}$.

A formula like $a_{n+1} = 3a_n + 1$ is a **recurrence relation**; $a_n = \frac{3^{n+1}-1}{2}$ is an **explicit** formula, which is also **closed**: it contains no $\sum$ or $\prod$ over a varying number of terms. For example $f(n) = \sum_{i=1}^{n} i$ is explicit but not closed, while $\frac{n(n+1)}{2}$ is closed. Induction often turns a recurrence into a closed formula.

**Sets.** A finite **set** is an unordered collection of distinct objects ($\{1, 3, 2\} = \{2, 1, 3\}$); if repetitions are allowed, it is a **multiset**. $B \subseteq A$ means every element of $B$ lies in $A$. Since enumerative statements usually depend only on the size of a set, we use the canonical $n$-element set $[n] := \{1, 2, \ldots, n\}$.

**Theorem 2.4**

For every positive integer $n$, $[n]$ has $2^n$ subsets.

**Proof.** $[1]$ has the two subsets $\emptyset, \{1\}$. Assume the claim for $n$. Subsets of $[n+1]$ not containing $n+1$ are exactly the subsets of $[n]$: $2^n$ of them. Those containing $n+1$ are $\{n+1\} \cup T$ with $T \subseteq [n]$ arbitrary: another $2^n$. Altogether $2^{n+1}$. ■

**Pitfalls**

The initial step cannot be skipped: "$2n+1$ is even" has a valid induction step ($2(n+1)+1-(2n+1) = 2$) but no true initial case. More subtly, "all horses have the same colour" can be "proved" as follows. Among $n+1$ horses, the first $n$ and the last $n$ are each monochromatic, and they overlap, so all $n+1$ are. But for $n = 1$ the two groups (first horse, last horse) are disjoint, so the step from 1 to 2 fails. The induction step must be valid for *every* $n$ from the initial value on.

## 2.2 Strong induction

**Motivation.** Some recurrences express $a_{n+1}$ through all earlier terms, so knowing only $a_n$ is not enough. Strong induction lets the induction step use the statement for *all* smaller values; it is no stronger logically, but it is often the convenient form.

**Example 2.5**

Let $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + \cdots + a_n + n + 1$ for $n \geq 0$. Then $a_n = 2^n - 1$.
*Solution.* True for $n = 0$. If $a_k = 2^k - 1$ for all $k \leq n$, then

$$a_{n+1} = \sum_{k=0}^{n} (2^k - 1) + n + 1 = (2^{n+1} - 1) - (n+1) + n + 1 = 2^{n+1} - 1.$$

**The method (strong induction).**

1. *Initial step:* prove the statement for the smallest $n$ for which it is defined.
2. *Induction step:* prove that the statement for all integers $< n+1$ implies it for $n+1$.

*Validity.* If $n+1$ were the smallest failure, the statement would hold for all smaller integers (by minimality and the initial step), and the induction step would give it for $n+1$. From now on $\mathbb{N}$ denotes the set of nonnegative integers.

**Example 2.6**

Let $f(0)=1$, $f(1)=2$ and $f(n+1)=f(n-1)+2f(n)$ for $n \geq 1$. Then $f(n) \leq 3^n$ for all $n \in \mathbb{N}$.
*Solution.* True for $n=0,1$. If it holds for all values $\leq n$ ($n \geq 1$), then $f(n+1) \leq 3^{n-1}+2\cdot 3^n = 7\cdot 3^{n-1} < 3^{n+1}$. Both $f(n-1) \leq 3^{n-1}$ and $f(n) \leq 3^n$ are used, so the strong hypothesis is needed.

**Variations**

It is often convenient to shift the parameter (assume the statement for $n-1$, prove it for $n$), or to induct over even or odd integers only; these variants are equally valid and are used in later chapters.

## 2.3 Selected exercises

**Motivation.** In these book exercises the difficulty is not the induction itself but choosing the right statement and the right object to remove in the induction step.

**Exercise 2.1 (Iterated differences of a polynomial)**

For a polynomial $p$ let $\Delta p(n) = p(n) - p(n-1)$, and let $\Delta^k p$ be the result of applying $\Delta$ $k$ times ($\Delta^0 p = p$). Prove that if $p$ has degree $d$, then $\Delta^d p$ is a constant.

**Solution.** We prove the sharper claim: if $p$ has degree $d$ and leading coefficient $a \neq 0$, then $\Delta^d p$ is the constant $d!\,a$.

*Step 1 (one difference lowers the degree).* For $j \geq 1$, the binomial theorem (Theorem 4.1) gives

$$n^j - (n-1)^j = n^j - \sum_{i=0}^{j} \binom{j}{i} n^i (-1)^{j-i} = j\, n^{j-1} + (\text{terms of degree} \leq j-2),$$

and for $j=0$ the difference is 0. Writing $p(n) = \sum_{j=0}^{d} c_j n^j$ with $c_d = a$ and applying this to each term,

$$\Delta p(n) = \sum_{j=1}^{d} c_j (n^j - (n-1)^j) = d\,a\,n^{d-1} + (\text{terms of degree} \leq d-2).$$

So for $d \geq 1$, $\Delta p$ has degree $d-1$ and leading coefficient $d\,a$.

*Step 2 (induction on $d$).* For $d=0$, $\Delta^0 p = p$ is the constant $a = 0!\,a$. Let $d \geq 1$ and assume the claim for degree $d-1$. By Step 1, $q = \Delta p$ has degree $d-1$ and leading coefficient $da$, so by the induction hypothesis $\Delta^{d-1} q = (d-1)! \cdot da = d!\,a$. Since $\Delta^d p = \Delta^{d-1}(\Delta p) = \Delta^{d-1} q$, the claim holds for $d$. ∎

**Exercise 2.2 (A player who lists everybody)**

At a tennis tournament every two players played each other exactly once. Afterwards each player listed the names of those he defeated and of those defeated by someone he defeated.

Prove that some player listed the names of everybody else.

**Solution.** Say that $x$ *covers* $y$ if $x$ beat $y$, or $x$ beat some $z$ who beat $y$. We prove by induction on the number $n \geq 1$ of players that some player covers all others.

*Step 1 (initial step).* For $n = 1$ there is nobody else to list, and the claim is vacuous.

*Step 2 (induction step).* Let $n \geq 1$, consider a tournament with $n+1$ players and fix a player $X$. Among the other $n$ players (with the games between them), the induction hypothesis gives a player $K$ who covers all of them. If $K$ beat $X$, or $K$ beat someone who beat $X$, then $K$ also covers $X$ and we are done.

*Step 3 (the remaining case).* Otherwise $X$ beat $K$, and $X$ beat every player $z$ beaten by $K$ (else $z$ beat $X$ and $K$ would cover $X$). Now take any $y \neq X$. If $y = K$, then $X$ beat $y$. If $K$ beat $y$, then $X$ beat $y$. If $K$ beat some $z$ who beat $y$, then $X$ beat $z$ and $z$ beat $y$. In every case $X$ covers $y$, so $X$ lists everybody else. ∎

## Exercise 2.15 (Arithmetic and geometric means)

Prove by induction that for all nonnegative reals $a_1, \ldots, a_n$,

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

**Solution.** Equivalently, $a_1 \cdots a_n \leq A^n$ where $A$ is the arithmetic mean. We induct on $n$; for $n = 1$ there is equality.

*Step 1 (setting up).* Let $n \geq 1$, assume the claim for $n$ numbers, and let $a_1, \ldots, a_{n+1} \geq 0$ have mean $A$. The claim does not depend on the order, so we may assume $a_n = \min_i a_i$ and $a_{n+1} = \max_i a_i$; then $a_n \leq A \leq a_{n+1}$.

*Step 2 (replacing two numbers).* Put $b = a_n + a_{n+1} - A$. Then $b \geq a_n \geq 0$, and

$$A\,b - a_n a_{n+1} = (A - a_n)(a_{n+1} - A) \geq 0, \qquad \text{so} \qquad a_n a_{n+1} \leq A\,b.$$

*Step 3 (applying the hypothesis).* The $n$ numbers $a_1, \ldots, a_{n-1}, b$ are nonnegative with sum $(n+1)A - A = nA$, i.e. with mean $A$. By the induction hypothesis $a_1 \cdots a_{n-1}\, b \leq A^n$. Hence

$$a_1 \cdots a_{n-1} a_n a_{n+1} \leq a_1 \cdots a_{n-1} \cdot A\,b \leq A^n \cdot A = A^{n+1}.$$

∎

## Exercise 2.17 (An L-shape of 1's)

(a) Let $m, n \geq 2$ and let $A$ be an $m \times n$ matrix with entries 0 and 1, at least $m + n - 1$ of which are 1. Prove that some entry 1 has another 1 in its row and another 1 in its column.
(b) Does the statement remain true with only $m + n - 2$ entries 1?

**Solution.**

(a) Suppose no entry 1 has the required property. Then every 1 is alone in its row or alone in its column. Let $R$ be the set of 1's alone in their row, and $C$ the set of the remaining 1's; each element of $C$ is alone in its column.

*Step 1 (counting).* Distinct elements of $R$ lie in distinct rows and distinct elements of $C$ in distinct columns, so $|R| \leq m$ and $|C| \leq n$. Since $|R| + |C| \geq m + n - 1$, either $|C| = n$ or $|R| = m$.

*Step 2 (case $|C| = n$).* Every column contains an element of $C$, which is the only 1 of that column. So every 1 of $A$ lies in $C$, and $R = \emptyset$. Then $n = |C| \geq m + n - 1$ gives $m \leq 1$, a contradiction.

*Step 3 (case $|R| = m$).* Symmetrically, every row contains exactly one 1, all 1's lie in $R$, $C = \emptyset$, and $m \geq m + n - 1$ gives $n \leq 1$, a contradiction.

(b) No. Put 1's at the positions $(1,1), \ldots, (1, n-1)$ of the first row and $(2, n), \ldots, (m, n)$ of the last column: $m + n - 2$ ones. A 1 in the first row is the only 1 of its column (columns $1, \ldots, n-1$ have no other 1), and a 1 in the last column is the only 1 of its row (rows $2, \ldots, m$). So no 1 has the required property. ■